Answer Key - Cybersecurity - Undergraduate Level Assessment 21 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** C
   Options: A) Private-key system; B) Shared-key system; C) Public-key system; D) All of them
   Note: Correct option: C - Public-key system

2. **Answer:** C
   Options: A) overwriting cryptographic keys in memory; B) a kind of code injection; C) a read outside bounds of a buffer; D) a format string attack
   Note: Correct option: C - a read outside bounds of a buffer

3. **Answer:** C
   Options: A) Physical layer; B) Data-link Layer; C) Session layer; D) Presentation layer
   Note: Correct option: C - Session layer

4. **Answer:** A
   Options: A) 160 bits; B) 512 bits; C) 628 bits; D) 820 bits
   Note: Correct option: A - 160 bits

5. **Answer:** B
   Options: A) Installing and configuring a IDS that can read the IP header; B) Comparing the TTL values of the actual and spoofed addresses; C) Implementing a firewall to the network; D) Identify all TCP sessions that are initiated but does not complete successfully
   Note: Correct option: B - Comparing the TTL values of the actual and spoofed addresses

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'Authentication'.

7. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

8. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: '128'.

9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

10. **Answer:** False
    Options: A) True; B) False
    Note: LLM-generated false statement. Correct answer: 'backdoor'.

Long Form Response

11. **Answer:** def find_Element(arr,ranges,rotations,index) : | return arr[index]
    Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** def count_list(input_list): | return len(input_list)
    Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key